

Artificial Intelligence

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence.

Machine Learning is a subset of AI.

Deep Learning is a subset of Machine Learning.

Generative AI can create text, images, code, audio and video.

Large Language Models (LLMs) include:

- Gemini
- GPT
- Claude

Retrieval-Augmented Generation (RAG) combines retrieval with generation.

Embeddings convert text into vectors.

Vector Databases store embeddings.

ChromaDB is an open-source vector database.

Pinecone is a managed cloud vector database.

Semantic Search finds information based on meaning instead of exact keywords.

Chunking breaks large documents into smaller meaningful pieces.

Metadata helps filter documents.